Exercise 4.3

1. (i) 15 kg. (ii) ₹ 36 2. (i) Direct proportion (ii) k = 5 (iii) C = 50
3. 90 months 4. 12 minutes 5. ₹ 140 6. 3 7. 300 litres

Challenge Problems

8. ₹ 1155 9. 35 minutes 10. 81 11. 6 days 12. 10 days 13. 5 days

GEOMETRY

Exercise 5.1

1. The pairs of adjacent angle are ∠ABG and ∠GBC, ∠BCF and ∠FCE, ∠FCE and ∠ECD, ∠ACF and ∠*FCE* , ∠ACF and ∠FCD.
2. 65° 3. 86° 4. (i) 108° (ii) 46° (iii) 30°
5. The other angle is also a right angle. 6. 30° , 120°, 210° 7. 63°
8. Adjacent angles: ∠PQU and ∠PQT, ∠PQT and ∠TQS, ∠TQS and ∠SQR, ∠SQR and ∠RQU, ∠RQU and ∠PQU (all possible answers).

   Vertically opposite angles: ∠PQU and ∠TQR, ∠PQT and ∠UQR.
9. 120° 10. 105° 11. 45°, 135° 12. (i) 125° (ii) 55°

Objective type questions

13. (iii) one common arm, one common vertex, no common interior
14. (iii) linear pair 15. (iv) equal in measure
16. (i) 360° 17. (ii) 80°

Exercise 5.2

1. (i) The angles are exterior angles on the same side of the transversal
   (ii) The angles are alternate exterior angles
   (iii) The angles are corresponding angles
   (iv) The angles are interior angles on the same side of the transversal
   (v) The angles are alternate interior angles
   (vi) The angles are corresponding angles
2. (i) 35° (ii) 65° (iii) 145° (iv) 135° (v)90°
3. (i) 28° (ii) 58° (iii) 123° (iv) 108°
4. (i) 149° (ii) 45° (iii) 101° (iv) 158°
5. (i) 42° (ii) 8° (iii) 2° (iv) 15°
6. (i) 55° (ii) 35°
7. (i) No. Since, interior angles on the same side of the transversal are supplementary.
   (ii) No. Since Corresponding angles are equal.
9. Minimum number of angles is 1. Using the concept of linear pair of angles, we can find one more angle and by the concepts of corresponding angles and alternate angles (interior and exterior) we could find all other angles.

Answers.indd 128 10/12/2022 18:28:11